

ETP formalization benchmark · story vs literal vs two-stage at scale (no-think) · vacuous laws excluded

Equational Theories Project implications rendered per each run's form, formalized back by each model over OpenRouter, and graded syntactically (checkform.py). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

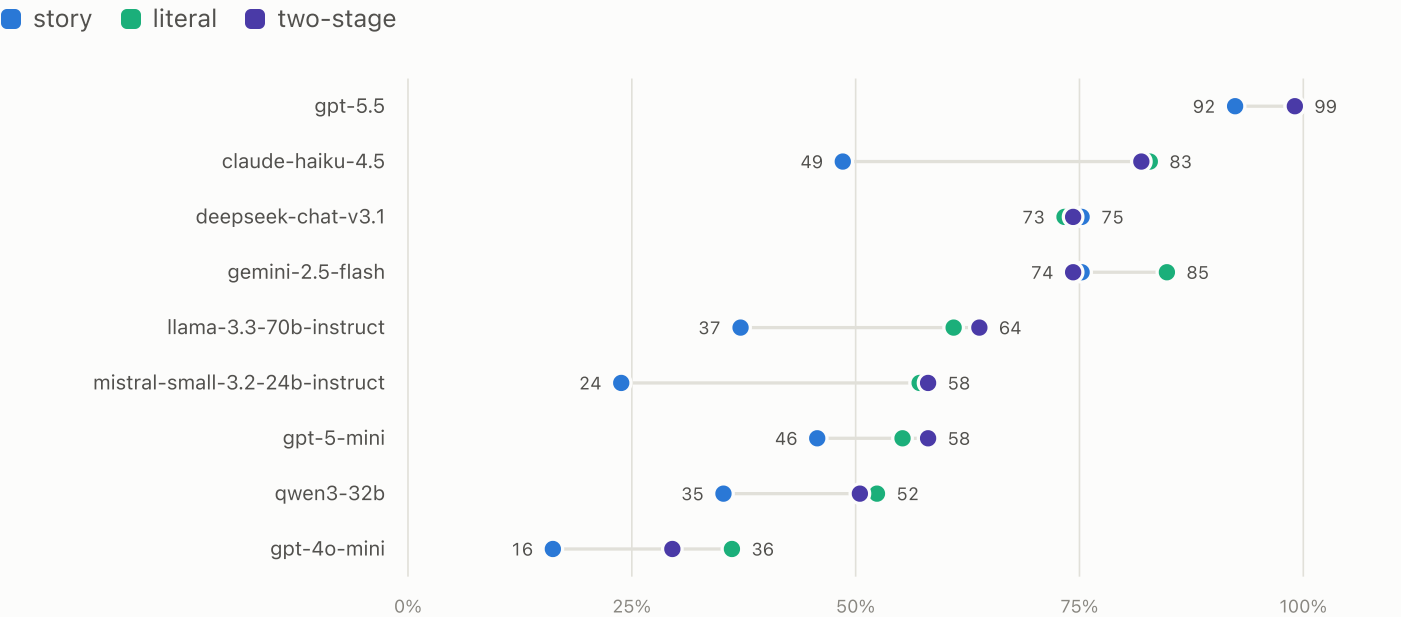
results/no-vacuous/run-strat20-s0-literal-think-off · results/no-vacuous/run-strat20-s0-story-think-off · results/no-vacuous/run-strat20-s0-think-off

Runs	Models	Stories	Graded calls	Overall correct
3	9	105	2,835	61%

105 pairs, 20 per operation-count bin 1–8 · seed 0 · story vs literal vs two-stage · no-think

FIGURE 1 Correct rate by model and form — no-think · stratified 105

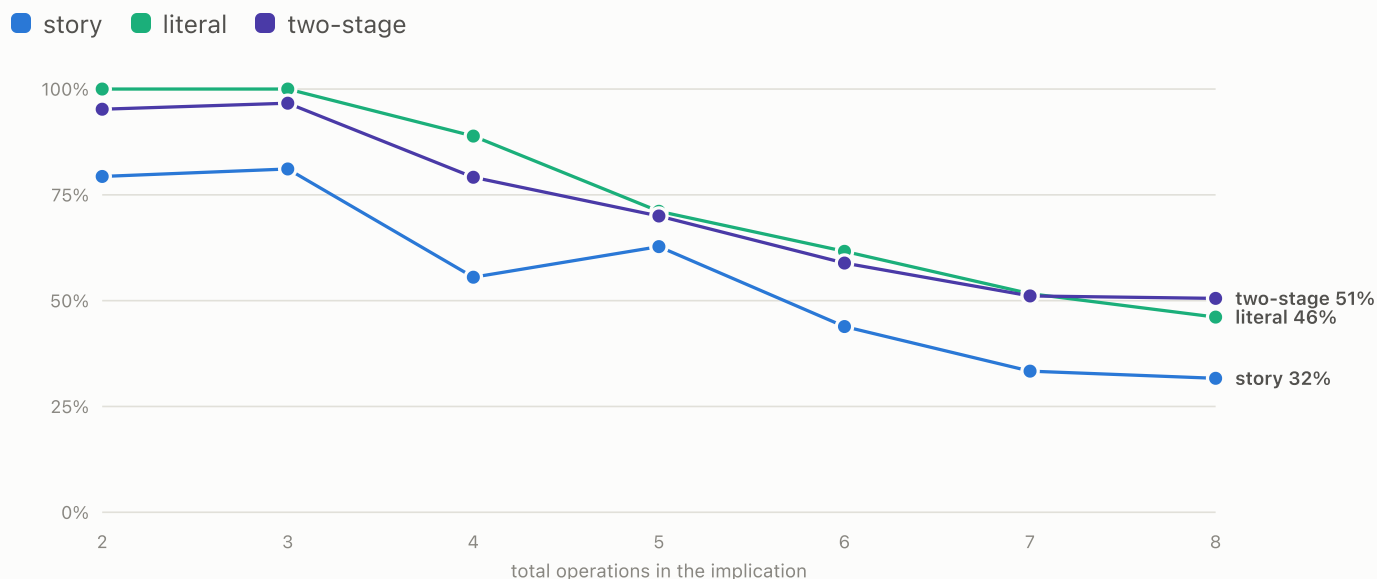
The same pairs rendered per arm; each row compares one model across the forms under the no-think regime. Sample: 105 pairs, 20 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: correct rate per model and form

FIGURE 2 **Accuracy vs operation count — no-think · stratified 105**

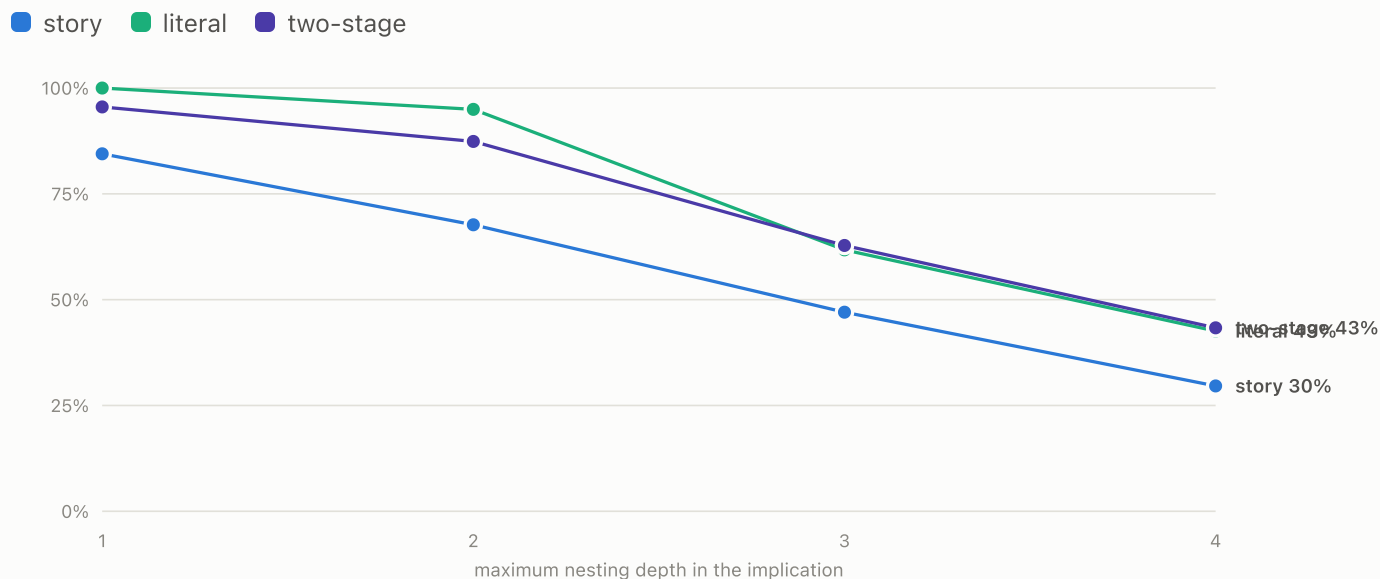
Correct rate pooled over all models, by the implication's total operation count (both equations combined); one line per form, no-think regime. Sample: 105 pairs, 20 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: pooled correct rate per operation-count bin

FIGURE 3 **Accuracy vs nesting depth — no-think · stratified 105**

Correct rate pooled over all models, by the deepest operation nesting across the four equation sides; one line per form, no-think regime. Sample: 105 pairs, 20 per total-operation bin 1–8 — the full complexity range, including trivial laws.

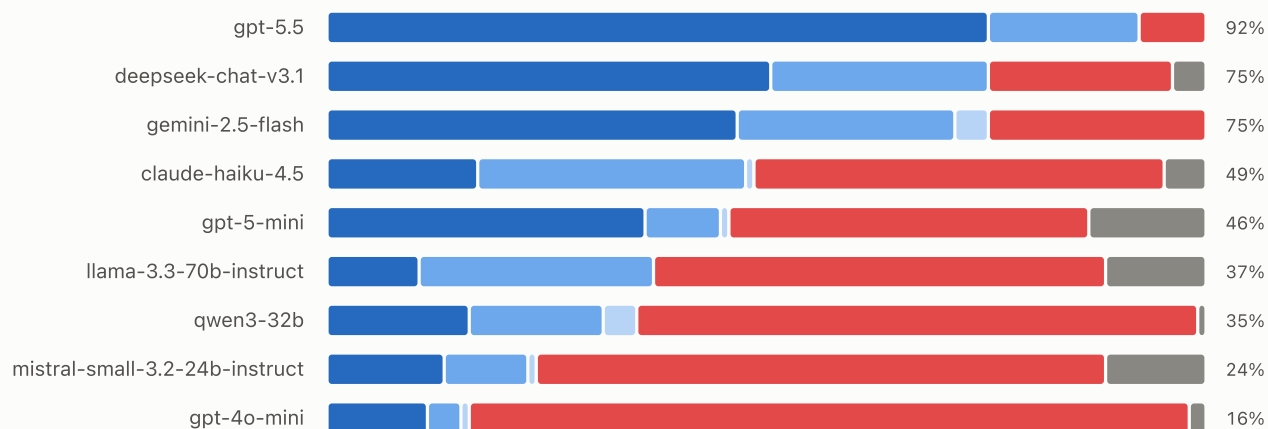


► Data: pooled correct rate per depth bin

FIGURE 4 **Verdict composition — story · no-think · stratified 105**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 105 pairs, 20 per total-operation bin 1–8 — the full complexity range, including trivial laws.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable

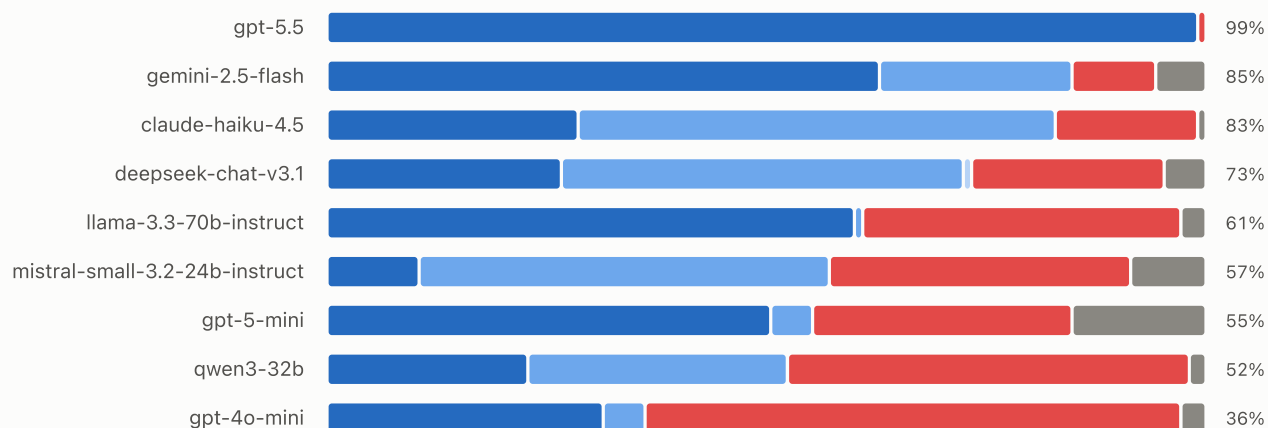


► Data: verdict counts per model

FIGURE 5 **Verdict composition — literal · no-think · stratified 105**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 105 pairs, 20 per total-operation bin 1–8 — the full complexity range, including trivial laws.

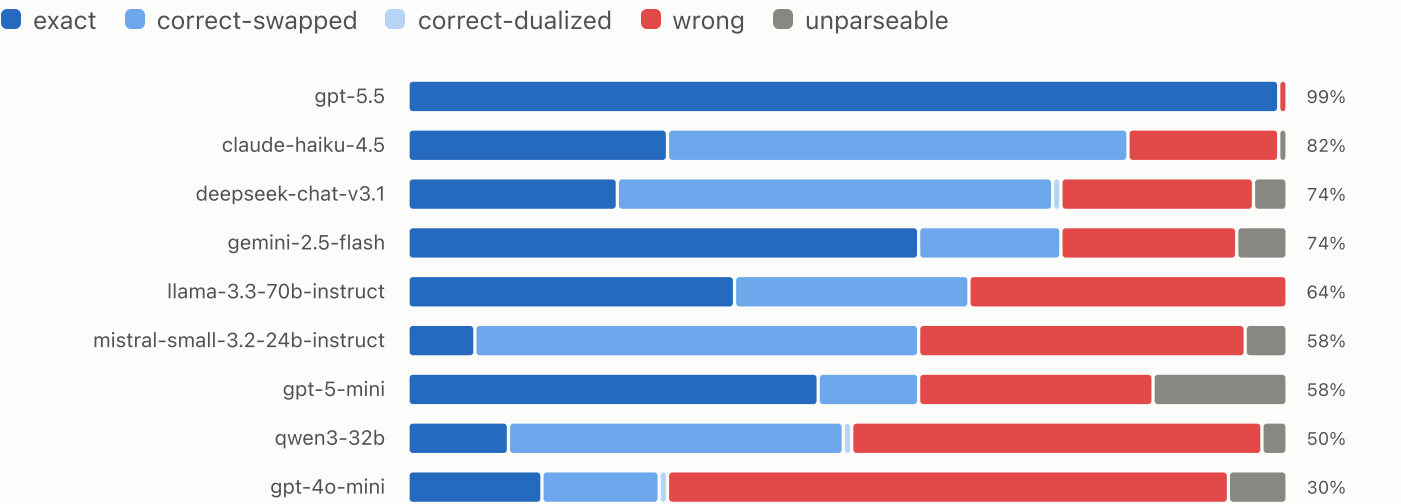
■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 6 **Verdict composition — two-stage · no-think · stratified 105**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 105 pairs, 20 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: verdict counts per model